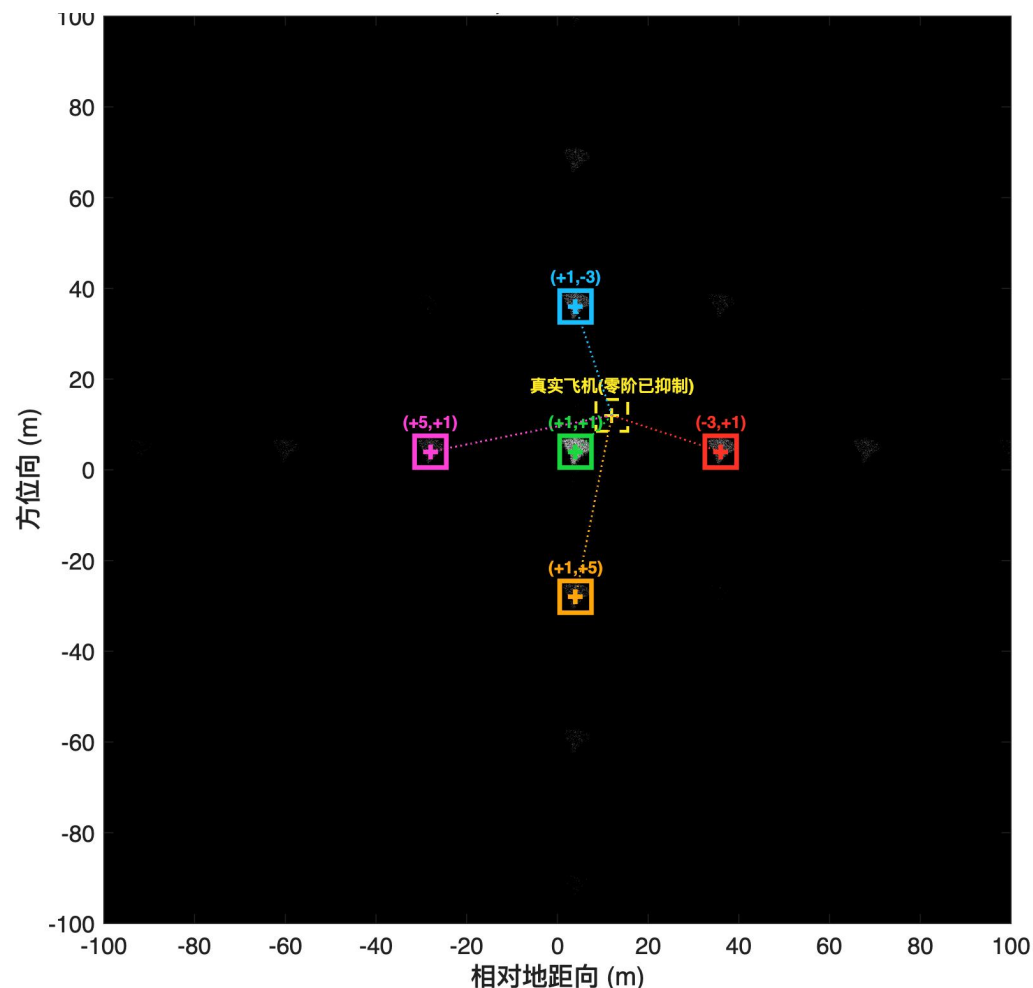
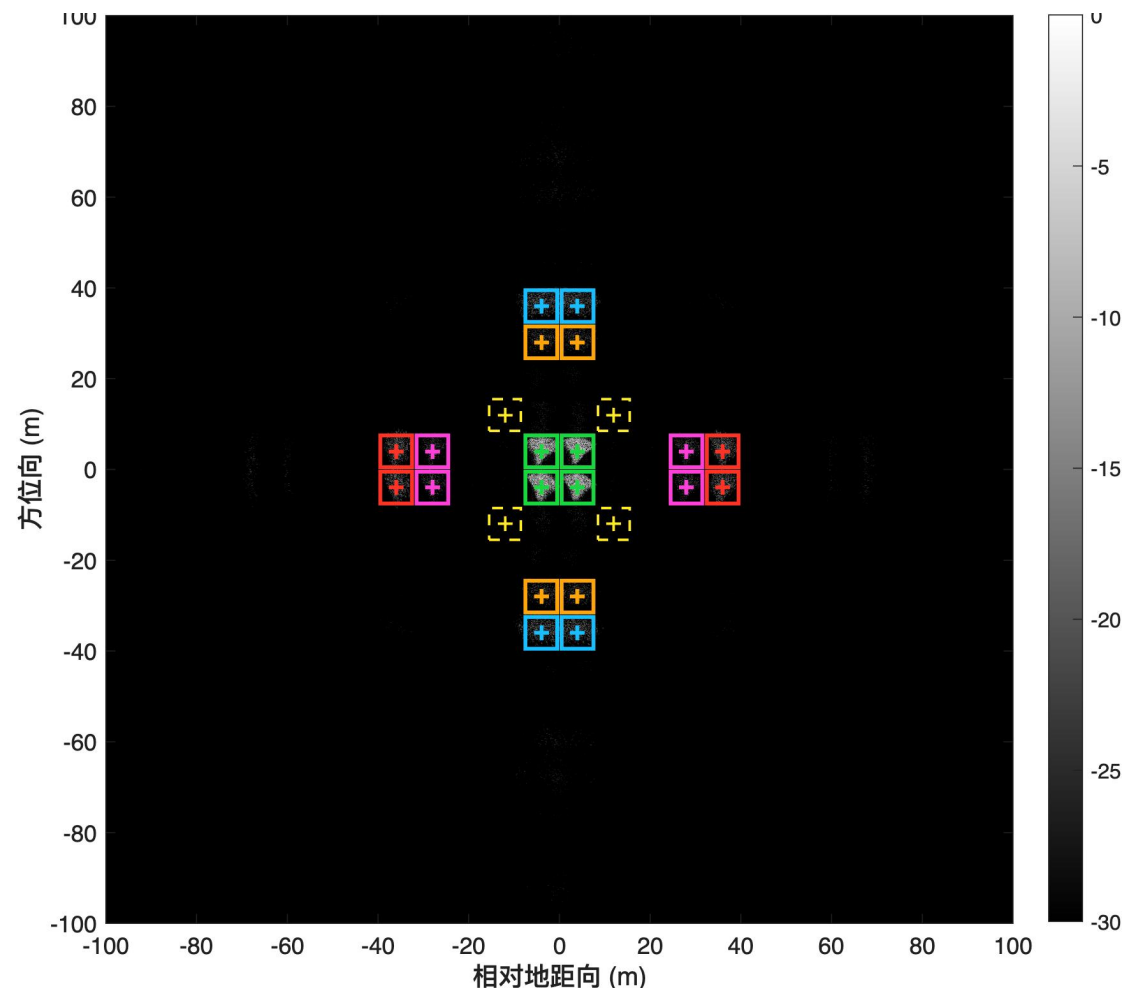


1 Matlab-飞机

- 成像结果中，会出现外侧十字的高阶谐波副像
在2bit，平衡编码时，主要是-3，1，5阶三种谐波



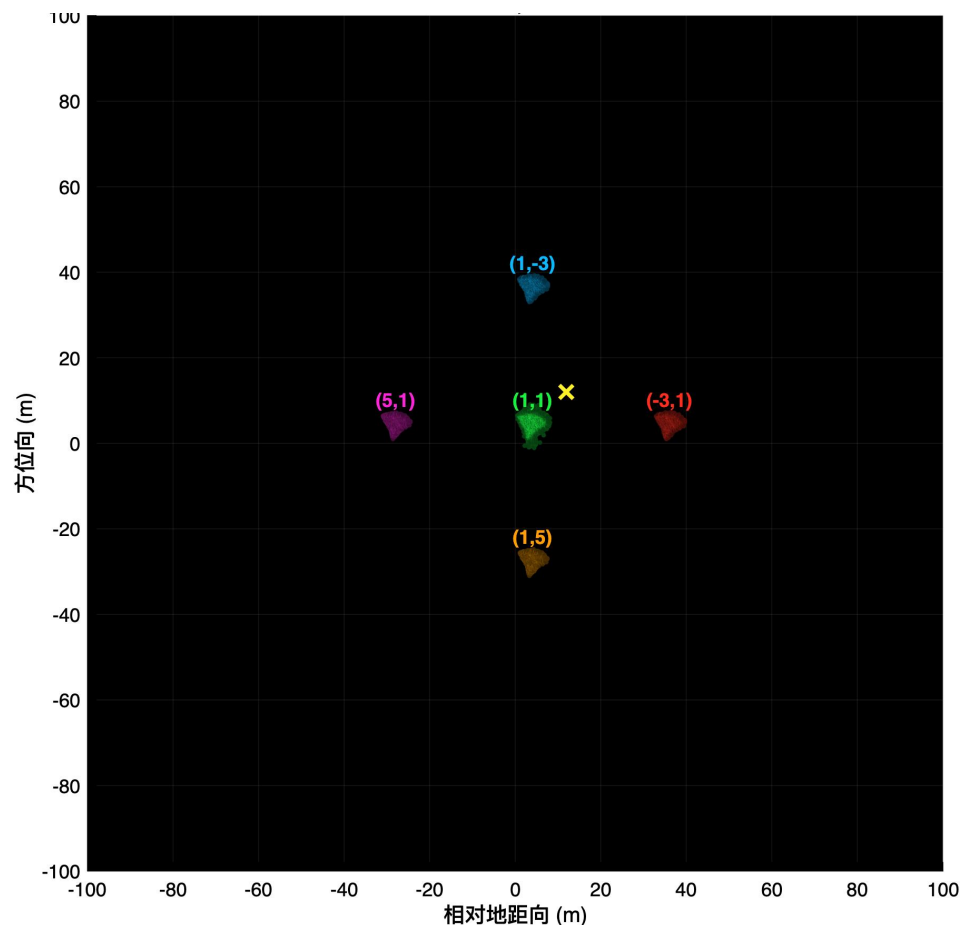
— 真实飞机位置 (零阶理论抑制) — (-3,1) 距离负三阶 — (1,-3) 方位负三阶
— (1,1) 中央目标 — (5,1) 距离正五阶 — (1,5) 方位正五阶



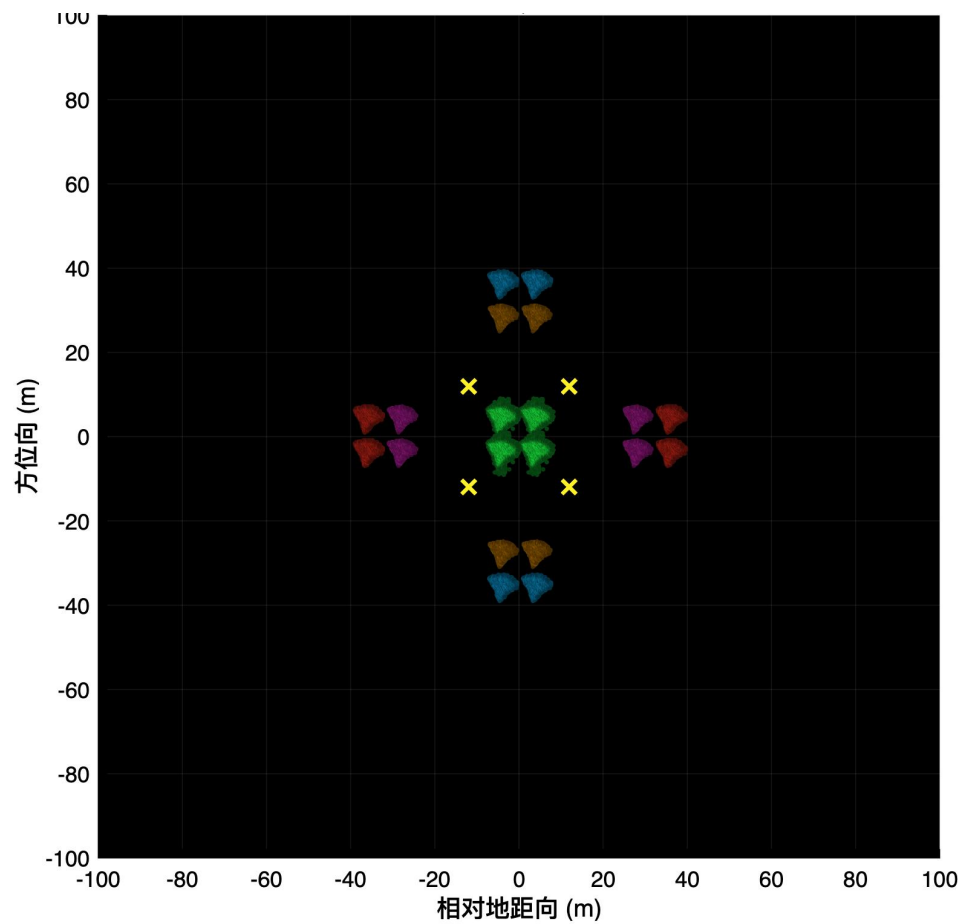
— 真实飞机位置 (零阶理论抑制) — (-3,1) 距离负三阶 — (1,-3) 方位负三阶
— (1,1) 中央目标 — (5,1) 距离正五阶 — (1,5) 方位正五阶

1 Matlab-飞机

- 成像结果中，会出现外侧十字的高阶谐波副像
在2bit编码时，频谱会保留 $n = \dots, -7, -3, +1, +5, +9, \dots$



× 真实飞机位置(零阶已抑制) (5,1) 距离正五阶十字副像
● (1,1) 期望2×2假目标 (1,-3) 方位负三阶十字副像
● (-3,1) 距离负三阶十字副像 (1,5) 方位正五阶十字副像



× 四架真实飞机位置(零阶已抑制) (5,1) 距离正五阶十字副像
● (1,1) 期望2×2假目标 (1,-3) 方位负三阶十字副像
● (-3,1) 距离负三阶十字副像 (1,5) 方位正五阶十字副像

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● 相关工作1：用的3-bit来改善谐波能量，模拟连续相位

Open Access Article

Variable Frequency Phase Modulation on Time-Modulated Metasurface for SAR Feature Reconstruction

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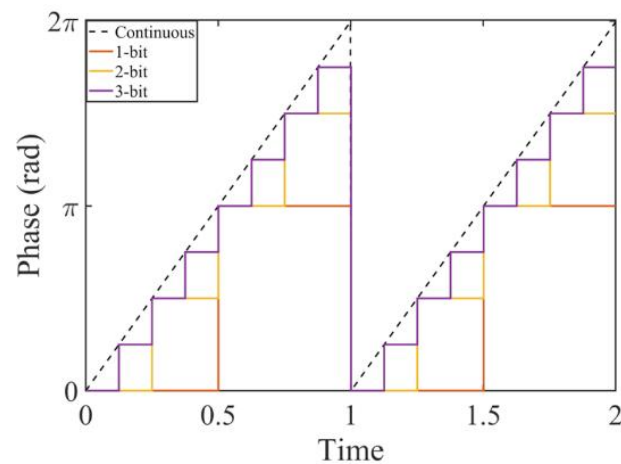
Highlights

What are the main findings?

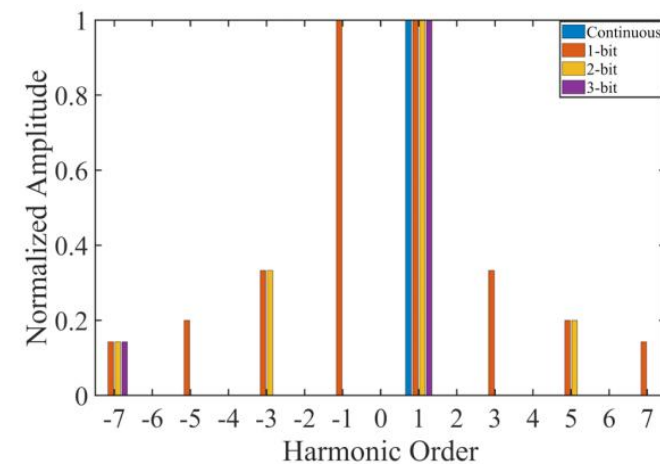
- A SAR feature reconstruction method based on variable frequency phase modulation is proposed, enabling a single metasurface to modulate multiple scattering centers in both range and azimuth dimensions.
- An analytical inverse model is established to link modulation parameters with the spatial coordinates and amplitudes of generated scattering centers, allowing for the simultaneous reconstruction of target features through independent design of modulation duration and frequency.

What are the implications of the main findings?

- This study marks the first systematic application of variable frequency-modulation techniques to SAR feature reconstruction, breaking through the inherent limitations of traditional one-to-one mapping. It provides a novel technical solution for achieving efficient, flexible, and high-fidelity simulation of complex target electromagnetic characteristics.
- An inverse analytical expression has been established between the target electromagnetic scattering characteristics and the time-domain modulation parameters of the metasurface. Through independent design of the modulation duration and modulation frequency, simultaneous reconstruction of the position and amplitude information of the generated scattering points is achieved.



(a)



(b)

Figure 3. Comparison of 1-bit, 2-bit, 3-bit, and continuous linear-phase modulation. (a) Modulation waveforms; (b) Harmonic order and normalized amplitude.

bit数越高，能量在1阶越集中

1 Matlab-飞机

- 相关工作1: 用的3-bit来改善谐波能量, 模拟连续相位

这篇文章里面有很多SAR成像图

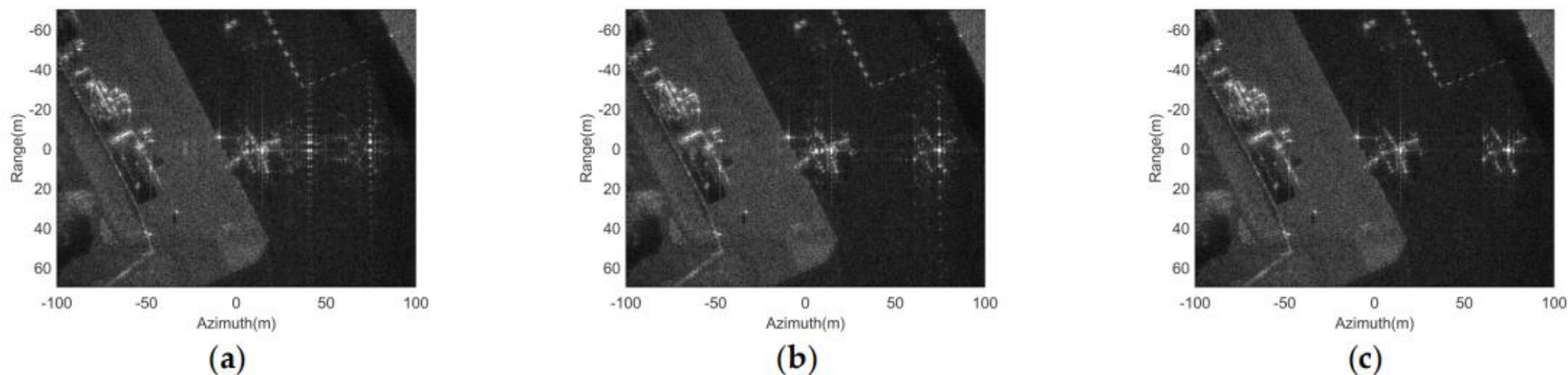


Figure 17. Comparison of SAR images of the reconstructed aircraft target and the original aircraft target under different phase quantization bits (a) 1-bit; (b) 2-bit; (c) continuous phase modulation.

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- 相关工作2：空时联合编码
- 为不同子区设计不同时间序列、起始相位和空间相位
- 各个分区之间是异步的

1. 时间延迟给k阶谐波添加k倍相位

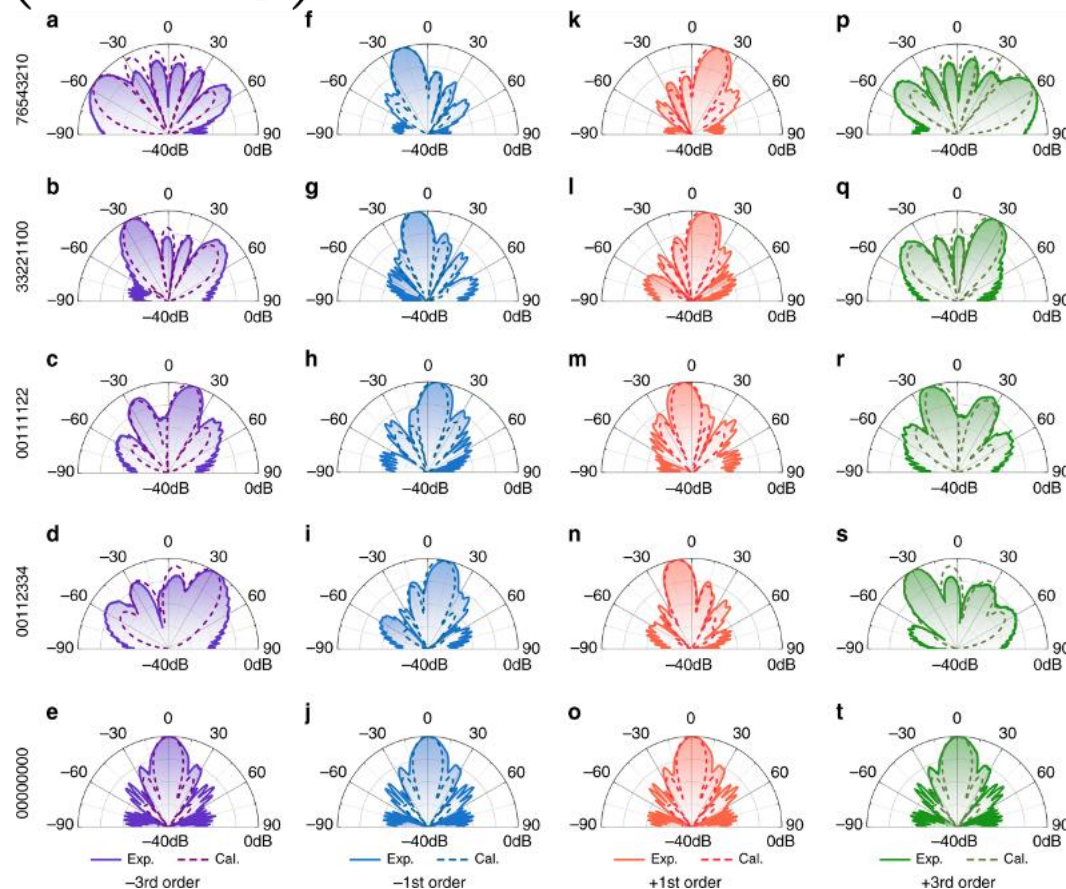
$$E_r(\omega) = 2\pi A \cos \frac{\phi_2 - \phi_1}{2} e^{j\frac{\phi_2 + \phi_1}{2}} E_i(\omega) + \sum_{m=-\infty}^{+\infty} \frac{4A}{2m-1} \sin \frac{\phi_2 - \phi_1}{2} e^{j[\frac{\phi_2 + \phi_1}{2} - (2m-1)\omega_0 t_0]} E_i[\omega - (2m-1)\omega_0] \quad (7)$$

Specifically, the amplitude and phase of the k^{th} order harmonic are rewritten as

$$E_r(k\omega_0 + \omega_c) = |A| \angle \Phi = \begin{cases} |A \cos \frac{\phi_2 - \phi_1}{2}| e^{j\{\frac{\phi_2 + \phi_1}{2} + \pi[\varepsilon(\frac{\phi_2 - \phi_1}{2}) + \varepsilon(\frac{\phi_1 - \phi_2}{2})\]}} & k = 0 \\ \frac{|2A|}{k\pi} \sin \frac{\phi_2 - \phi_1}{2} e^{j\{\frac{\phi_2 + \phi_1}{2} - \pi[\varepsilon(\phi_2 - \phi_1) + \varepsilon(k)] - k\omega_0 t_0\}} & k = \pm 1, \pm 3, \pm 5 \dots \\ 0 & k = \pm 2, \pm 4, \pm 6 \dots \end{cases} \quad (8)$$

The extra phase shift provides new degrees of freedom to control the propagation features of high-order harmonics by breaking the constraint between their amplitude and phase, making it possible to realize independent regulation of both parameters, which is hard to achieve using conventional methods. In fact, Eq. (8) offers a promising route to tailor the response of reflected

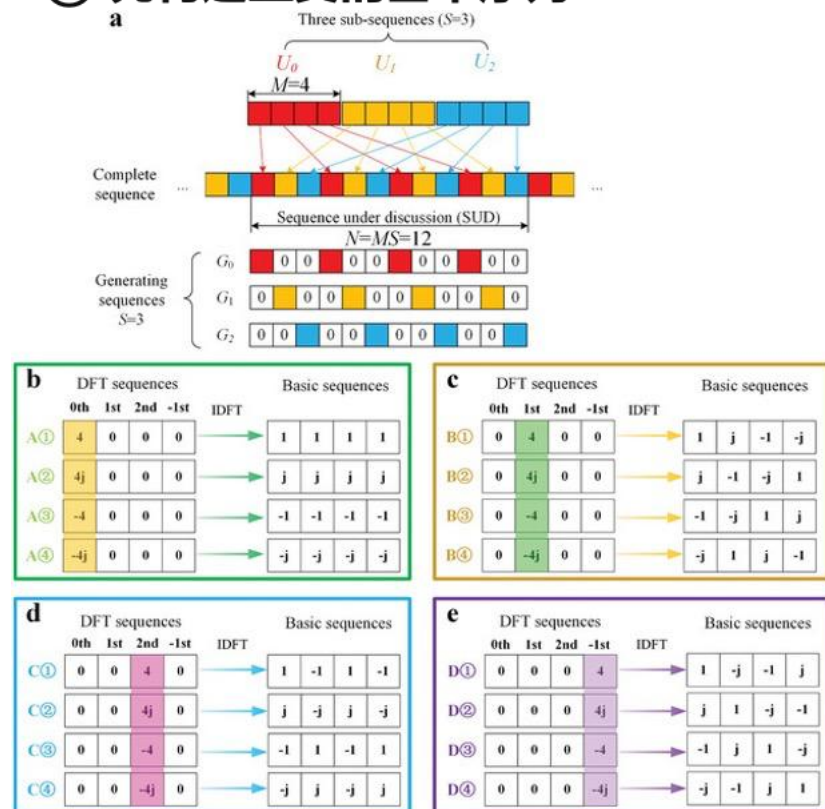
$$\Gamma(t - t_0) \implies e^{-jk\omega_0 t_0}$$



1 Matlab-飞机

2. 交织子序列把2-bit时间码变成可设计的谐波向量

① 先构造正交的基本序列



$S=3$
 $M=4$
 $\alpha = -30^\circ$

i : index of sub-sequence
 k : harmonic order

Complex plane
0th
+1st

② 再把各子序列贡献做复数矢量叠加

Sub-sequences:

U_0 : $A①[1, 1, 1, 1]$
 U_1 : $A①[1, 1, 1, 1]$
 U_2 : $A①[1, 1, 1, 1]$

Sub-sequences:

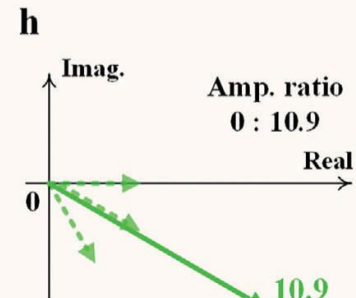
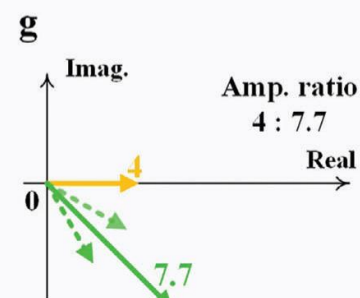
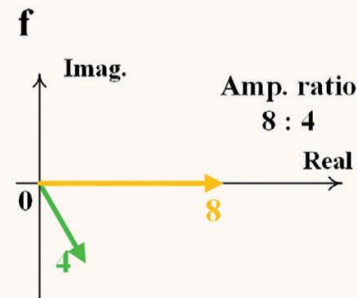
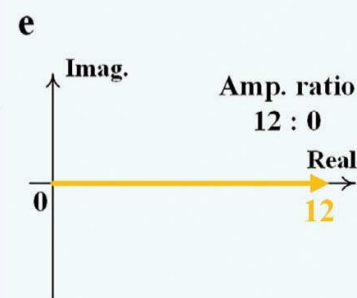
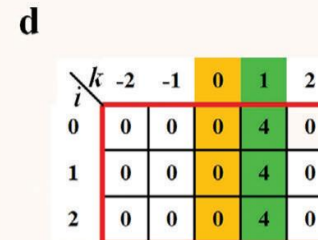
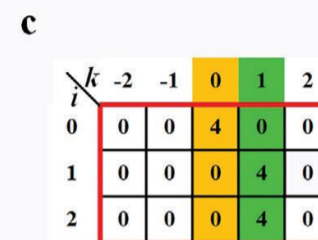
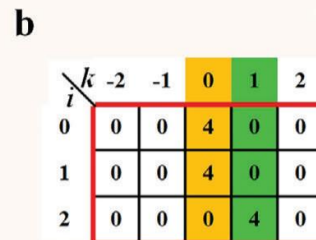
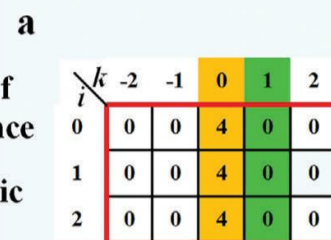
U_0 : $A①[1, 1, 1, 1]$
 U_1 : $A①[1, 1, 1, 1]$
 U_2 : $B①[1, j, -1, -j]$

Sub-sequences:

U_0 : $A①[1, 1, 1, 1]$
 U_1 : $B①[1, j, -1, -j]$
 U_2 : $B①[1, j, -1, -j]$

Sub-sequences:

U_0 : $B①[1, j, -1, -j]$
 U_1 : $B①[1, j, -1, -j]$
 U_2 : $B①[1, j, -1, -j]$



1 Matlab-飞机

2. 交织子序列把2-bit时间码变成可设计的谐波向量

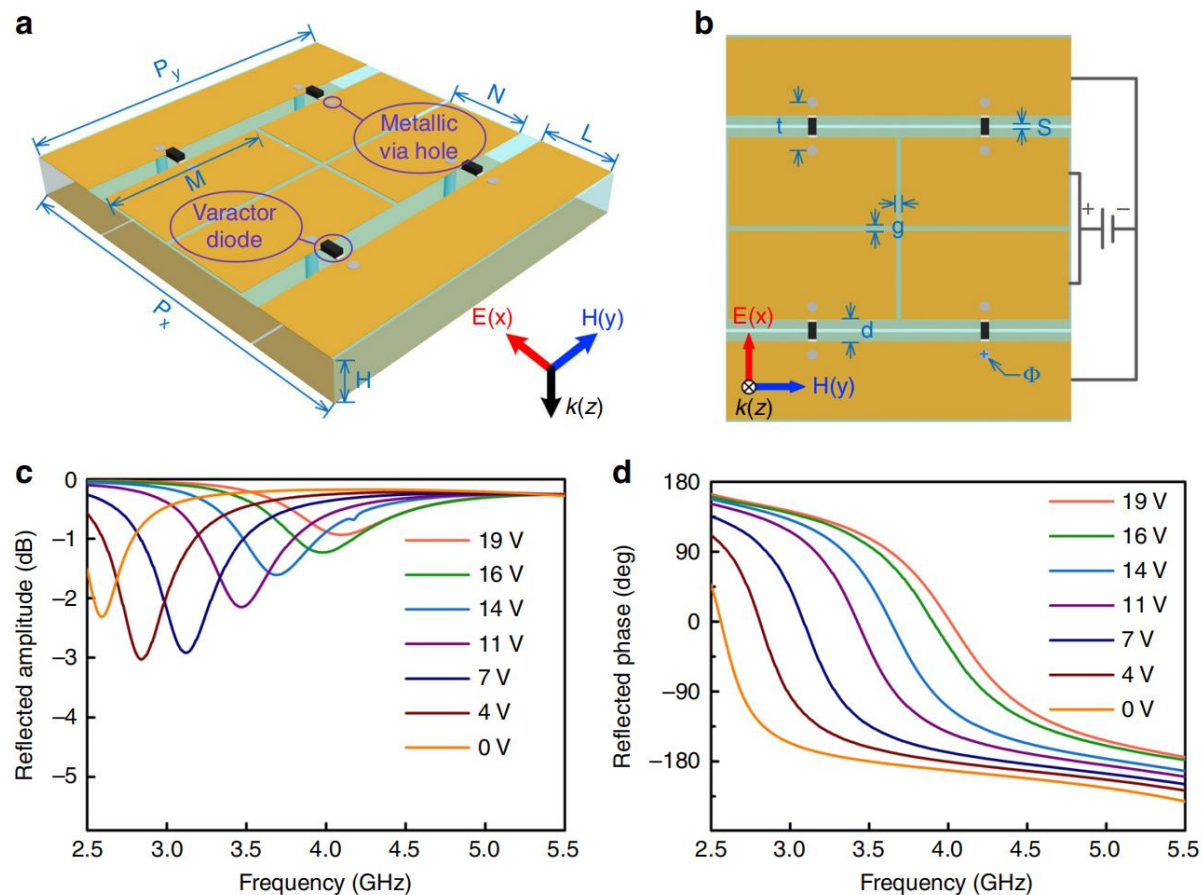
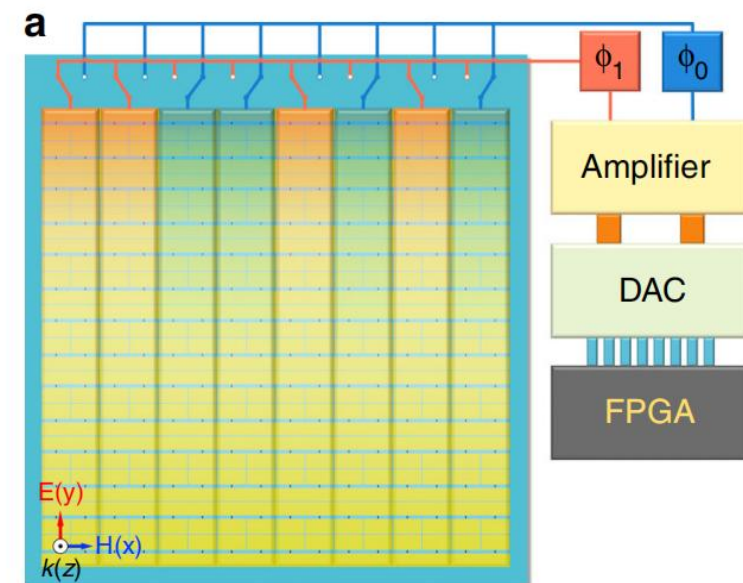


Fig. 2 Illustration of the time-domain digital coding meta-atom. **a, b** Perspective and front views of the time-domain digital coding meta-atom. **c, d** The simulated reflection amplitude (**c**) and phase (**d**) spectra under different bias voltages. The dimensions of the meta-atom are $P_x = 37$ mm, $P_y = 33$ mm, $H = 4$ mm, $M = 16$ mm, $N = 8.45$ mm, $L = 7.45$ mm, $g = 0.5$ mm, $d = 2.1$ mm, $t = 4.6$ mm, $\Phi = 1$ mm, $s = 0.3$ mm



1 Matlab-飞机

思路

一个局部4×4超单元： $P_{ij}=R_i \times A_j$

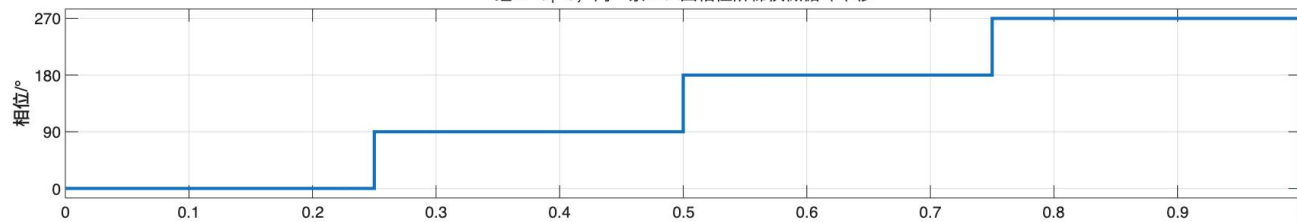
每格仍是单极化2-bit四相位单元，不是固定相位块

方位向时延组 A_j

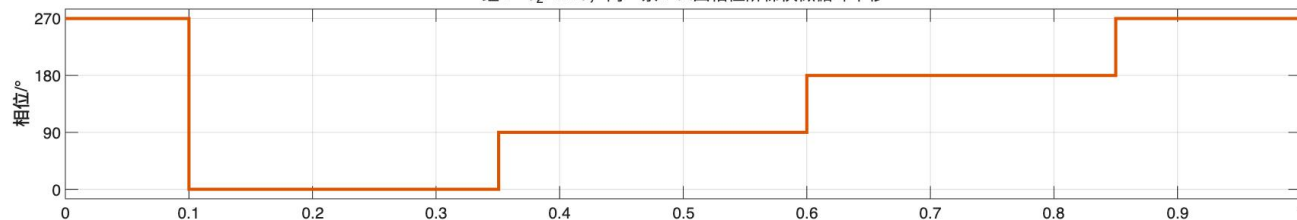
	A_1	A_2	A_3	A_4
R_1	P_{11} $d_r=0$ $d_a=0$	P_{12} $d_r=0$ $d_a=T/10$	P_{13} $d_r=0$ $d_a=5T/6$	P_{14} $d_r=0$ $d_a=14T/15$
R_2	P_{21} $d_r=T/10$ $d_a=0$	P_{22} $d_r=T/10$ $d_a=T/10$	P_{23} $d_r=T/10$ $d_a=5T/6$	P_{24} $d_r=T/10$ $d_a=14T/15$
R_3	P_{31} $d_r=5T/6$ $d_a=0$	P_{32} $d_r=5T/6$ $d_a=T/10$	P_{33} $d_r=5T/6$ $d_a=5T/6$	P_{34} $d_r=5T/6$ $d_a=14T/15$
R_4	P_{41} $d_r=14T/15$ $d_a=0$	P_{42} $d_r=14T/15$ $d_a=T/10$	P_{43} $d_r=14T/15$ $d_a=5T/6$	P_{44} $d_r=14T/15$ $d_a=14T/15$

四个距离组和四个方位组使用相同的延迟集合
一个 P_{ij} 同时选取第 i 条距离序列和第 j 条方位序列

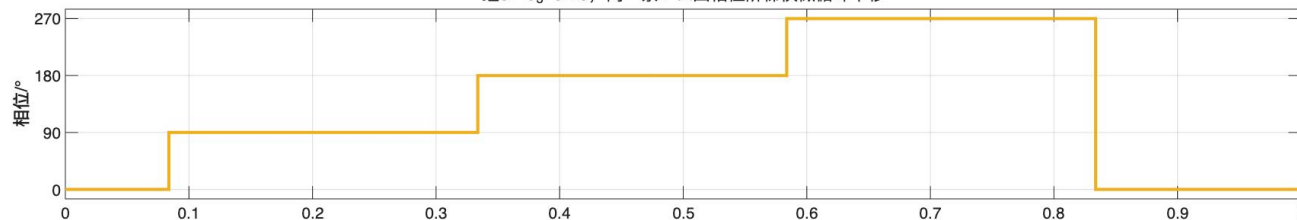
组1: $d_1=0$, 同一条2-bit四相位阶梯仅做循环平移



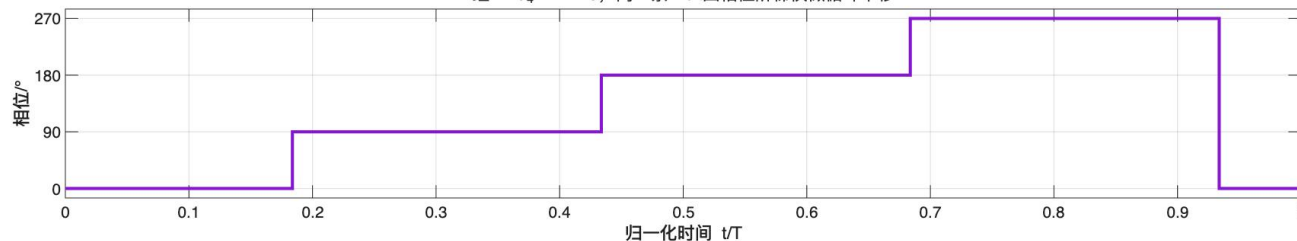
组2: $d_2=T/10$, 同一条2-bit四相位阶梯仅做循环平移



组3: $d_3=5T/6$, 同一条2-bit四相位阶梯仅做循环平移

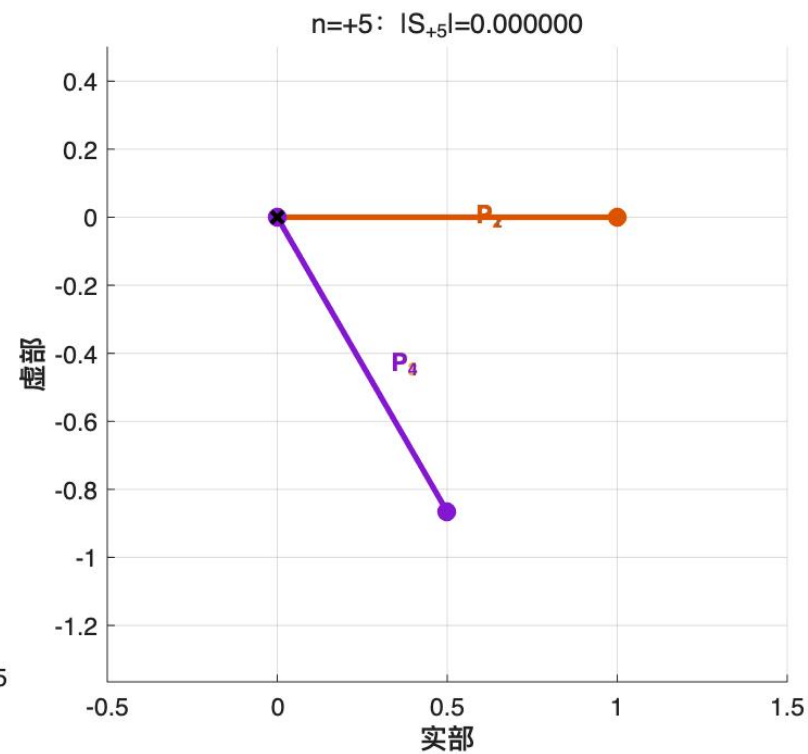
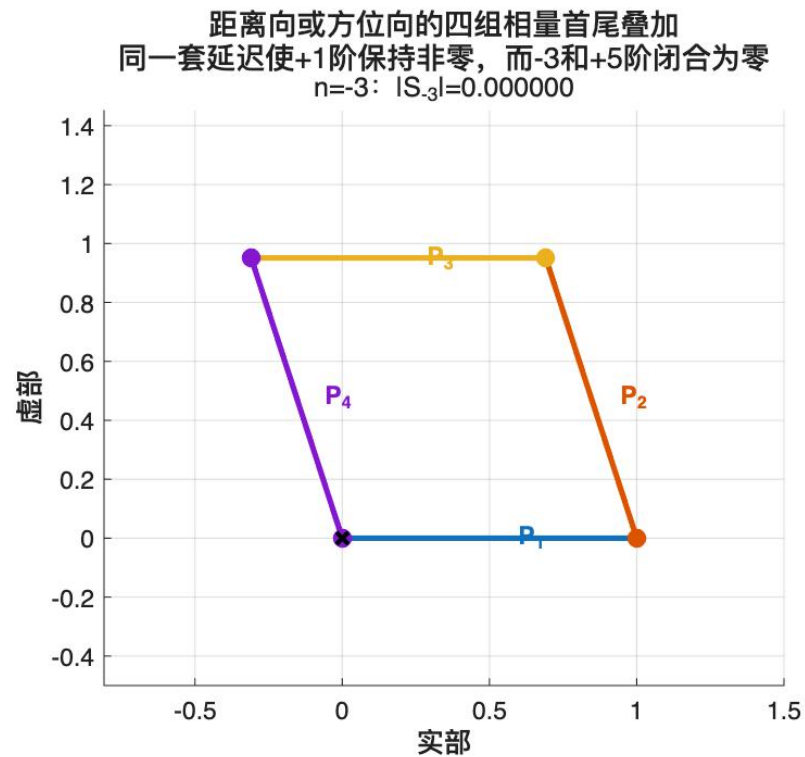
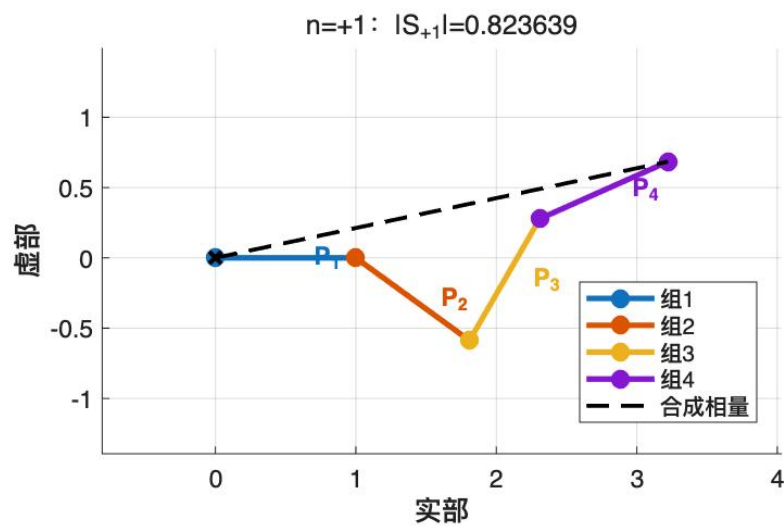


组4: $d_4=14T/15$, 同一条2-bit四相位阶梯仅做循环平移



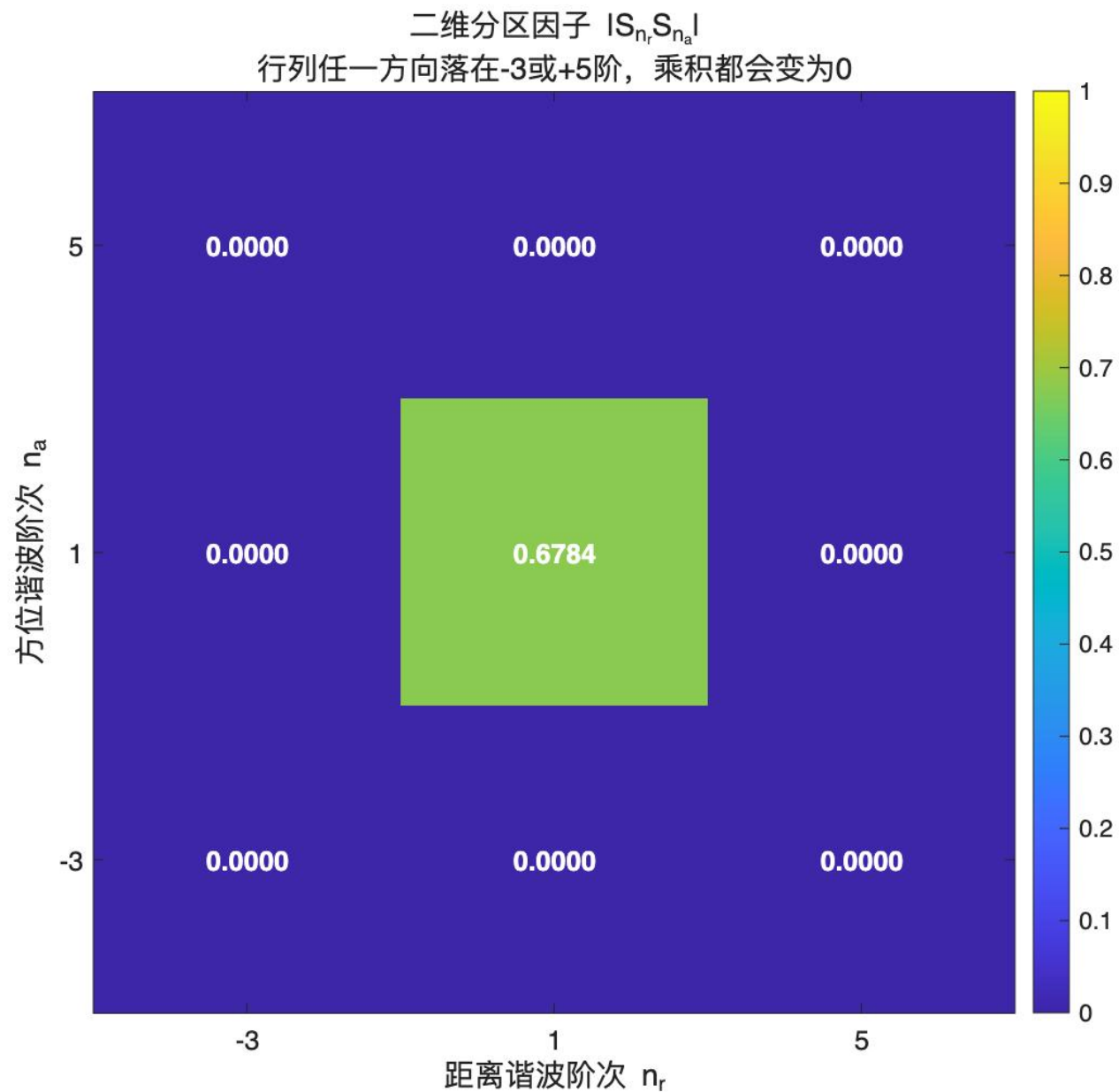
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思路



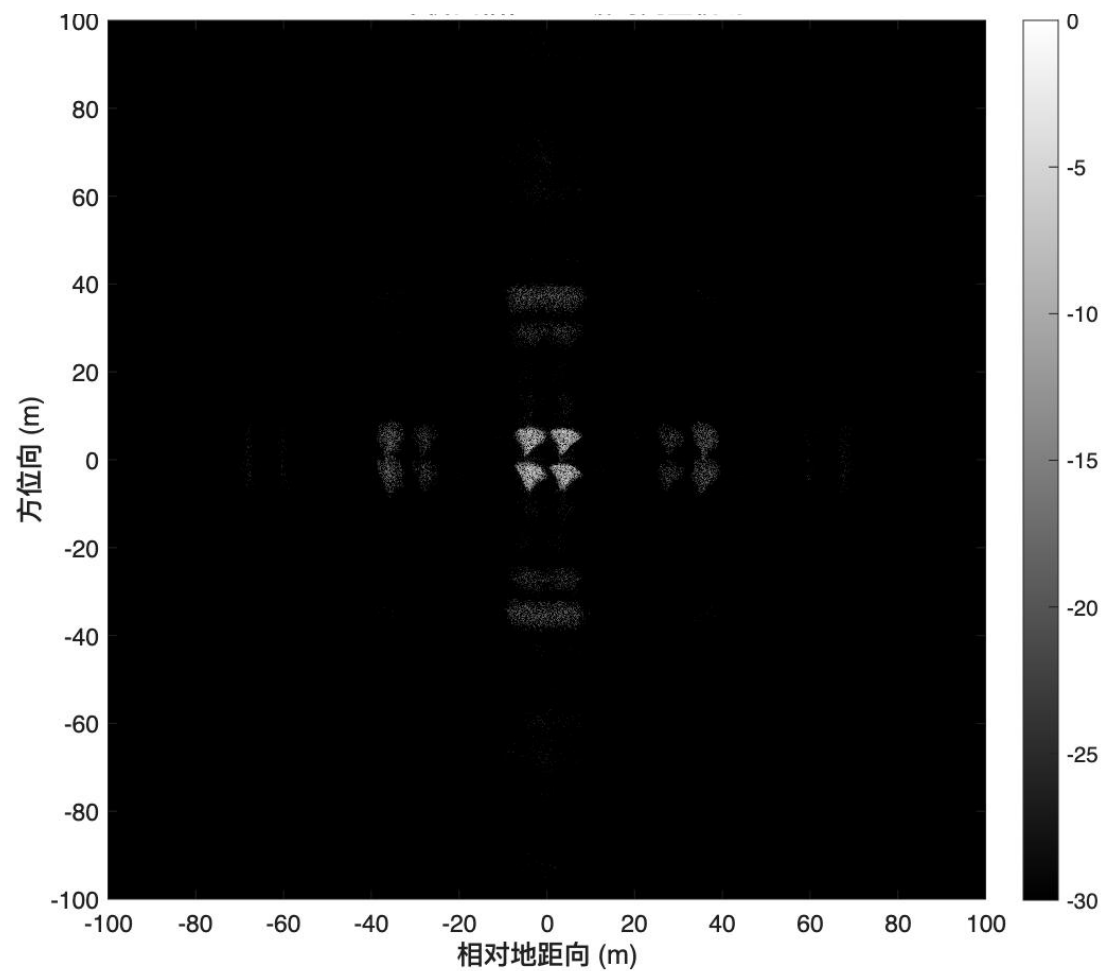
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思路

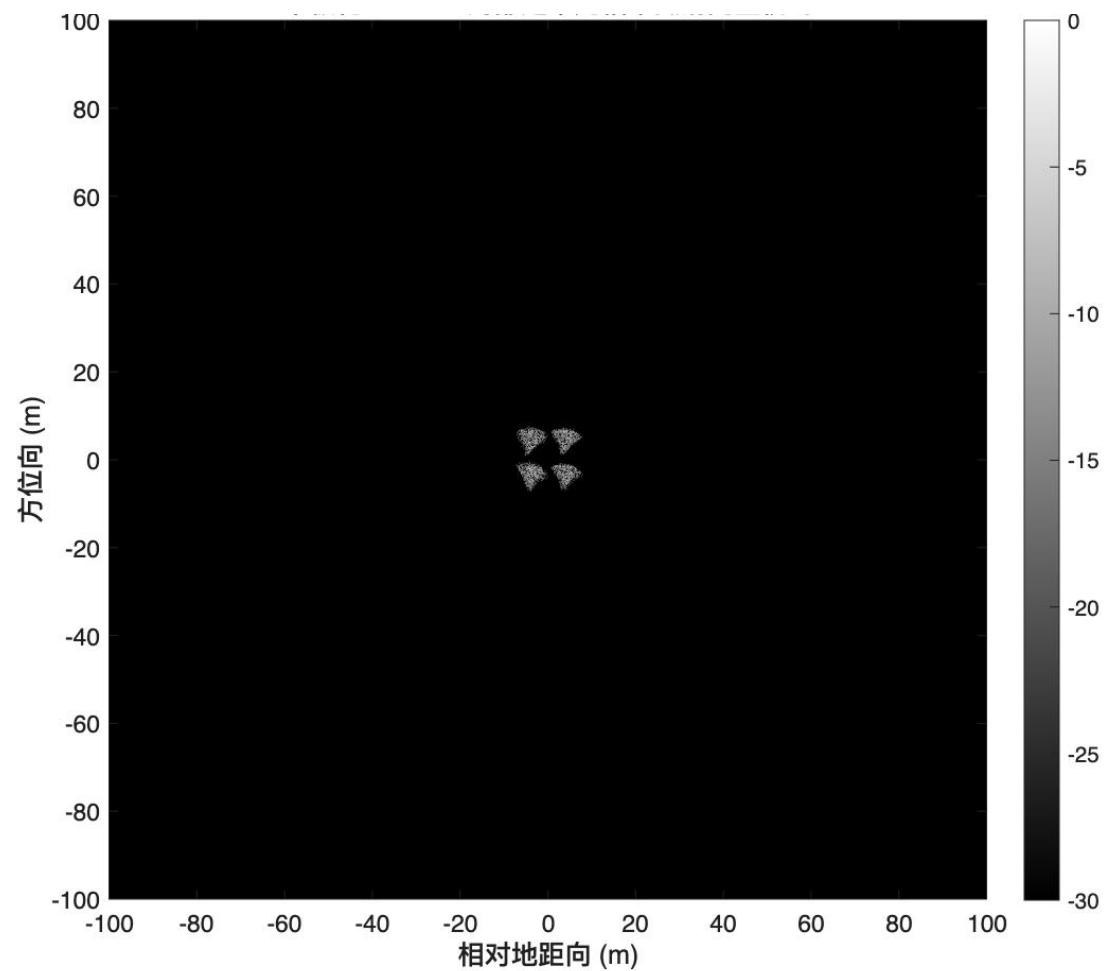


1 Matlab-飞机

思路



优化前



优化后